

PROJECT REPORT

Project Name- Agentic AI for Smart Event Management Operations

Abstract

EventOps AI is an enterprise-oriented event intelligence platform designed to consolidate critical event-management activities into a single operational control plane. The platform brings together registration intelligence, venue operations, speaker operations, schedule and conflict management, sponsorship intelligence, incident management, operational alerts, agent orchestration and platform operations.

The system combines structured operational data with AI-assisted workflows to provide a unified view of event health, operational risk, recommendations and domain-specific insights. It is designed to support event teams in monitoring activities, entering operational records, identifying important signals and responding to issues from one dashboard.

The project is demonstrated using an AI Conference 2026 scenario. It provides dedicated interfaces for operational data entry and monitoring while demonstrating how specialized AI agents can support different event-management functions.

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1. Introduction

Managing a modern conference or large-scale event requires coordination among multiple teams. Registration teams manage attendees and check-ins, venue teams manage rooms and capacity, event coordinators handle speakers and schedules, sponsorship teams monitor commercial partnerships, and operations teams respond to incidents and alerts.

When these activities are managed through disconnected tools, obtaining a real-time view of the complete event becomes difficult. EventOps AI addresses this challenge by presenting the major operational areas through one centralized dashboard.

The platform follows a modular architecture in which each business domain has a dedicated interface and supporting backend operations. The result is a practical control plane that combines event operations with AI-assisted decision support.

2. Problem Statement

Conventional event operations may depend on spreadsheets, separate registration tools, room schedules, manual communication and independent incident records. This can lead to:

- Limited visibility across operational teams.
- Difficulty monitoring registration and live check-in status.
- Inefficient venue allocation based on capacity and equipment requirements.
- Difficulty identifying suitable speakers based on expertise and availability.
- Limited visibility into sponsor performance and engagement.
- Manual identification, prioritization and escalation of incidents.
- Lack of a unified view of event health, risk and recommendations.
- Limited visibility into the technical health of the supporting platform.

3. Objectives

The primary objective is to develop a unified event intelligence platform that supports event teams through centralized operational management and AI-assisted decision support.

- Create a centralized executive dashboard for event health and operational KPIs.
- Manage attendee registration and check-in tracking.
- Maintain venue information including capacity, equipment, status and utilization.
- Support expertise-based speaker search and speaker directory management.
- Provide schedule visibility and conflict detection for overlapping venue sessions.
- Track sponsorship contracts, payments, leads and engagement.
- Record incidents and support priority and escalation workflows.
- Display and resolve operational alerts.
- Provide an orchestration view for specialized AI agents.
- Monitor service health, latency, requests, errors and platform-readiness indicators.

4. Proposed Solution

EventOps AI provides a web-based control plane with a centralized navigation system and a dynamic operational workspace. Users can move between event domains without leaving the platform.

The Executive Dashboard provides the overall event picture by consolidating registration, venue, sponsor and incident information. Individual modules then provide deeper operational capabilities such as data entry, search, analysis, recommendations and incident handling.

The modular approach also makes the platform easier to extend. New event workflows or external integrations can be added without redesigning the entire user experience.

5. System Architecture

The project follows a client-server architecture. The frontend is responsible for the interactive dashboard and sends API requests to the backend. The backend processes event operations, manages data access and provides agent-related endpoints. The data layer stores operational entities such as attendees, venues, speakers, sessions, sponsors and incidents.

Layer	Responsibility
Frontend	Interactive dashboard, navigation, forms, KPI cards, tables, analytics and agent views.
API / Backend	Event operations, records, agent endpoints, alerts, incidents and monitoring.
Data Layer	Persistence of event-management records.
Intelligence Layer	Event intelligence, venue recommendations, speaker matching, sponsorship analysis and incident/risk workflows.
Monitoring Layer	Service status, uptime, request count, errors, error rate, latency and resource indicators.

The current development configuration expects the backend service to be available on port 5000 while the frontend runs separately.

6. Major Modules and Features

6.1 Executive Dashboard

Provides event health and risk indicators, registration, venue, sponsor and alert KPIs, recommendations, AI insights, agent runtime status and milestone analytics.

6.2 Registration Intelligence

Supports attendee creation, attendee search and check-in tracking. Summary information includes total attendees, checked-in attendees and pending arrivals.

6.3 Venue Operations

Manages venue name, type, capacity, location, equipment, status and utilization. The Venue Agent can support venue recommendations based on operational requirements.

6.4 Speaker Operations

Provides expertise-based speaker search and a speaker directory containing name, role, organization, expertise, availability and rating.

6.5 Schedule & Conflict Detection

Provides session, speaker, venue, time, attendee and status information. Backend validation supports conflict detection for overlapping sessions in the same venue.

6.6 Sponsorship Intelligence

Tracks sponsor name, tier, contract value, amount paid, leads generated and engagement. Sponsor analysis provides performance information and recommendations.

6.7 Incident Management

Records operational issues with severity, affected people, owner and description, supporting incident priority and escalation.

6.8 Operational Alerts

Displays active alerts, messages and sources and provides an action to resolve active alerts.

6.9 Agent Orchestration

Provides a common view of Registration, Venue, Speaker, Sponsorship and Incident agents, together with an orchestration pipeline and cross-domain intelligence.

6.10 Platform Operations

Displays service health, uptime, latency, request count, errors, error rate, memory usage and platform-readiness information.

7. AI Agent Layer

The platform uses specialized agents to separate responsibilities by operational domain.

Agent	Role
Registration Agent	Supports attendee-focused analytics and registration operations.

Venue Agent	Recommends suitable venues using capacity, equipment and utilization signals.
Speaker Agent	Searches for speakers according to expertise and returns matching results.
Sponsorship Agent	Analyzes sponsor performance and provides recommendations or action items.
Incident Agent	Supports risk analysis, priority and escalation for operational incidents.

The orchestration layer provides a common view of these agents, allowing domain-specific intelligence to be demonstrated through one operational dashboard.

8. Data Flow and Operational Workflow

A typical user interaction begins in the frontend. The user enters information or requests an analysis, the frontend sends an API request to the backend, and the backend validates and processes the request. The result is then returned to the frontend and displayed in the relevant module.

Example: Add Sponsor Workflow

- Open Sponsorship Intelligence.
- Select Add Sponsor.
- Enter sponsor information.
- Submit the form.
- Refresh the operational data.
- Verify the updated sponsor directory and KPIs.

Example: Incident Workflow

- Create an incident with title, severity, affected people, owner and description.
- Store the incident through the backend.
- Review priority/escalation information.
- Monitor the incident from Incident Management.

9. Technology Stack

Technology / Component	Purpose
React	Component-based frontend user interface.
Vite	Frontend development and build tooling.
Lucide React	Interface icons.
Backend API	Server-side event operations and intelligent workflows.
REST APIs	Communication between frontend and backend.
Database Layer	Persistence of operational records.
AI / Agent Workflows	Domain-specific recommendations, matching and analysis.
CSS	Dashboard layout and visual styling.

The final repository configuration should be used to specify the exact backend framework and database technology in a formal implementation submission

10. User Interface and Dashboard Design

The interface is designed as an operational dashboard rather than a simple data-entry application. The navigation provides quick access to each business domain, while the main workspace changes according to the selected module.

- Consistent KPI cards make important metrics easy to scan.
- Panels group related operational information.
- Tables provide structured records for attendees, venues, speakers, sponsors and incidents.
- Forms support direct entry of operational records.
- Agent actions are presented close to the related business data.
- The platform provides a live indicator and automatic refresh behavior.
- Backend connectivity errors are surfaced to the user when the service is unavailable.

11. Demonstration Data

The following sample records can be used during the project demonstration. They are demo values and can be replaced with actual event information.

Module	Sample Record 1	Sample Record 2
Registration	Aarav Sharma aarav.sharma@example.com IIT Delhi Student 18-25 Web	Priya Verma priya.verma@example.com TechNova Solutions Professional 26-35 Partner Portal
Venue	Innovation Hall Conference 500 Block A Projector, Wi-Fi, Stage Available 65%	AI Auditorium Auditorium 800 Main Campus Projector, Wi-Fi, Audio Available 40%
Speaker	Dr. Neha Kapoor Keynote Speaker AI Research Labs Artificial Intelligence Available 4.8	Rahul Mehta Technical Speaker TechNova Solutions Computer Vision Available 4.5
Sponsor	TCS Gold 500000 300000 150 leads 75%	Infosys Platinum 1000000 750000 320 leads 88%
Incident	Projector Failure in Hall A Medium 80 AV Support Team	Crowd Congestion at Gate 2 High 250 Operations Team

For incident entry, use the fields that are actually available in the running interface, such as title, severity, affected people, owner and description.

12. Testing and Validation

Test Area	Expected Validation
Frontend Startup	Dashboard loads when the backend service is running.
Backend Connectivity	Frontend API requests succeed when the backend is available on the configured port.
Registration	New attendee can be submitted and appears after refresh.
Check-in	Registered attendee can be marked as checked in.
Venue	New venue appears in venue inventory after creation.
Venue Agent	Recommendation request returns results when supported by backend data.
Speaker	New speaker appears in the speaker directory

	after creation.
Speaker Agent	Expertise search returns matching results.
Sponsor	New sponsor is created and sponsor KPIs update after refresh.
Sponsor Analysis	Analyse action returns sponsor performance information when supported.
Incident	New incident appears in the incident command center.
Alerts	Active alert can be resolved when an active alert is present.
Monitoring	Platform page displays service and request metrics returned by the backend.

13. Conclusion

EventOps AI demonstrates how an event-management platform can be organized around a unified intelligence layer. The project combines operational record management with AI-agent concepts, analytics, alerts and monitoring so that different event functions can be viewed from a single control plane.

The primary strength of the project is the integration of multiple event domains into one dashboard. Registration, venues, speakers, schedules, sponsorships and incidents contribute to a broader operational picture through KPIs, health indicators, risk signals and recommendations.

With further development, the platform can be extended toward real-time event intelligence, predictive safety monitoring, deeper integrations and production-scale deployment. The current implementation provides a practical foundation for demonstrating AI-assisted event operations and intelligent decision support.